Understanding the proof that $z(20) = 6$

A walkthrough of the cochromatic number of 20-vertex graphs (Erdős Problem #758)

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1 Part 1: What we are proving

Cochromatic number. For a graph G, a *cocolouring* splits the vertices into groups where each group is either a **clique** (everyone joined) or an **independent set** (nobody joined). Call such a group *homogeneous*. Then $z(G)$ is the fewest groups you need.

Quick sanity check: a 5-cycle. Any three of its vertices contain both an edge and a non-edge, so every homogeneous set has size at most 2 — you need at least $\lceil 5/2 \rceil = 3$ groups, and 3 works. So $z(C_5) = 3$.

The quantity we want is

$$z(20) \;=\; \max\{\, z(G) : G \text{ a graph on 20 vertices} \,\}.$$

That "max" means the proof splits in two, and both halves are genuinely needed:

	what it says	how you prove it
$z(20) \le 6$	*every* 20-vertex graph needs at most 6	a universal claim — hard
$z(20) \ge 6$	*some* 20-vertex graph needs 6	exhibit one graph — easy

> **Insight.** The asymmetry is the whole story. Lower bounds on a maximum are cheap: produce one example. Upper bounds are expensive: rule out 2^{190} possibilities. Almost all the machinery goes into the $\le$ direction.

2 Part 2: The lower bound (the easy half)

We need one 20-vertex graph that genuinely cannot be done in 5 groups.

The Paley graph on 17 vertices. Vertices are $0, \ldots, 16$. Join i and j when $i - j$ is a **quadratic residue** mod 17 — a nonzero perfect square mod 17. Those are $\{1, 2, 4, 8, 9, 13, 15, 16\}$.

The property that matters: **its largest clique is 3, and its largest independent set is 3.**

This is provable by hand, and worth seeing. The maps $x \mapsto ax + b$ with a a residue act transitively on edges (note $-1 \equiv 16$ is a residue, so $x \mapsto -x + 1$ swaps 0 and 1), so any hypothetical K_4 may be assumed to contain the edge $\{0, 1\}$. The common neighbours of 0 and 1 are $Q \cap (1 + Q) = \{2, 9, 16\}$, and their pairwise differences are 7 and 14 — neither a quadratic residue. So those three are pairwise non-adjacent, and no two can complete $\{0, 1\}$ to a K_4. Multiplying by a non-residue is an isomorphism to the complement, so there is no independent 4-set either.

Now the pigeonhole. Every homogeneous set has at most 3 vertices. To cover 17 vertices with groups of size at most 3 you need at least $\lceil 17/3 \rceil = 6$ groups. So $z(\text{Paley}(17)) \ge 6$.

Getting from 17 to 20. Add three isolated vertices. Why does that not help? Because of a general fact:

> If H sits inside G as an induced subgraph, then $z(H) \le z(G)$.

Why: take any cocolouring of G and delete the vertices not in H. Each group shrinks, and a subset of a clique is still a clique, a subset of an independent set still independent. So you get a valid cocolouring of H with no more groups.

So our 20-vertex graph contains Paley(17), hence needs at least 6. $z(20) \geq 6$**.**

And here is an explicit six-class cocolouring, so the value is exactly 6 (vertices $17, 18, 19$ isolated):

$$I_1 = \{0, 3, 6, 17, 18, 19\} \quad I_2 = \{1, 4, 7\} \quad I_3 = \{2, 9, 12\}$$
$$I_4 = \{5, 8, 11\} \quad I_5 = \{10, 13, 16\} \quad K_1 = \{14, 15\}$$

The first five are independent, the last is an edge. **The entire lower bound is therefore hand-checkable** — no solver is involved at any point.

3 Part 3: The upper bound — the target

We must show *every* 20-vertex graph G can be done in 6.

The strategy is subtraction. Suppose we could always find **two disjoint homogeneous 4-sets** A and B in G. Make them two groups. That uses 8 vertices and 2 groups. The remaining 12 vertices form some graph, and if every 12-vertex graph needs at most 4 groups, we are done:

$$2 \text{ groups} + 4 \text{ groups} = 6 \text{ groups}.$$

So the upper bound reduces to two claims:

- **(I)** Every 20-vertex graph contains two disjoint homogeneous 4-sets.
- **(II)** $z(12) \leq 4$.

> **Insight.** Notice what just happened: we replaced "partition 20 vertices into 6 homogeneous classes" — a statement about the whole structure — with "find two disjoint 4-sets". Two 4-sets is a much smaller object to hunt for. This is the move that makes everything else possible, and it is why an earlier 17-hour search failed: it was searching over *partitions* when it should have been reasoning about *4-sets*.

4 Part 4: Claim (I) — the heart of it

Step 1 — one 4-set is free. Ramsey's theorem says $R(4, 4) = 18$: any graph on 18 or more vertices contains 4 mutual friends or 4 mutual strangers. Since $20 \geq 18$, our G contains at least one homogeneous 4-set. Call it A.

Step 2 — assume A is a clique, without loss of generality. A might be an independent set instead. If so, flip G to its complement. Under complementation cliques become independent sets and vice versa, so *homogeneous sets are exactly preserved*, and so is the property "has two disjoint homogeneous 4-sets". Relabel so $A = \{0, 1, 2, 3\}$. (Part 9 examines this step in detail.)

Step 3 — the collapse. Suppose, for contradiction, that G has **no** two disjoint homogeneous 4-sets. Look at the other 16 vertices, call them B.

B cannot contain a homogeneous 4-set — any such set would live entirely in B, hence be disjoint from A, giving us the two disjoint sets we assumed do not exist.

So $G[B]$ is a 16-vertex graph with no K_4 and no independent 4-set. These are extremely rare objects, and this is the punchline:

> **There are exactly two such graphs, up to isomorphism.**

Step 4 — what is left to decide. So G must look like: the clique A on 4 vertices, one of exactly 2 fixed graphs on B, and then the $4 \times 16 = 64$ edges between them. Everything else is pinned down. Two problems of 2^{64} each — small enough to settle exhaustively.

Step 5 — the computation. For each of the two cores we ask a SAT solver: *is there any way to wire the 64 cross edges so that no two disjoint homogeneous 4-sets appear?* Both answers: **no** (unsatisfiable).

Therefore Step 3's assumption was false. **Every 20-vertex graph has two disjoint homogeneous 4-sets.**

5 Part 5: Claim (II) — $z(12) \leq 4$

Same shape of argument, one size down. Let G have 12 vertices.

- **If G has a homogeneous 4-set:** use it as one group. The remaining 8 vertices need at most $z(8) \leq 3$ groups. Total 4.
- **If G has none:** then G is a hom-4-free graph on 12 vertices. There are $1{,}449{,}166$ of these. Every single one was checked to be 4-cocolourable.

The cases are exhaustive, so $z(12) \leq 4$. And $z(8) \leq 3$ was checked over all $12{,}346$ graphs on 8 vertices — every graph, not a sample.

6 Part 6: Putting it together

$$\underbrace{\text{Ramsey}}_{\text{one 4-set}} \to \underbrace{\text{complement}}_{A \text{ is a clique}} \to \underbrace{\text{2 cores}}_{B \text{ pinned down}} \to \underbrace{\text{2 UNSATs}}_{\text{two disjoint 4-sets}} \to \underbrace{2 + z(12){=}4}_{z(20)\leq 6}$$

together with $z(20) \geq 6$ from Paley(17), gives $\boxed{z(20) = 6}$.

7 Part 7: Where a sceptic should push

The load-bearing joints, and how each was addressed:

1. **"Exactly two 16-vertex cores"** — a catalogue claim. Regenerated independently (Part 11); the level counts matched published values and level 18 came out empty, re-deriving $R(4,4) = 18$.
2. **The SAT encoding** — does it really capture "no two disjoint hom-4s"? Tested by generating 300 random cross-edge assignments per core, building the actual 20-vertex graph, computing directly whether it has two disjoint hom-4s, and confirming agreement with CNF satisfaction. 300/300 both times.
3. **The refutations** — checked by three independent checkers (a hand-written RUP checker, `drat-trim`, and `cake_lpr` whose core is formally verified), against CNFs regenerated from scratch, with deliberate wrong-proof controls that all correctly failed.
4. $z(12) \leq 4$ **and** $z(8) \leq 3$ — exhaustive, not sampled.
5. **The trusted remainder.** The proof checkers certify the *SAT layer* — that the CNFs given to them are unsatisfiable. They do not certify that the graph-to-CNF translation has the stated semantics, that the generator enumerates every isomorphism class, or that the cocolourability procedure searches exhaustively. Those encoding lemmas are supported by semantic spot-testing and by differential validation against published data, not proved. A correlated implementation error remains logically possible.

8 Part 8: The complement step — why the WLOG is free

This is the move that feels like sleight of hand, so let us be precise.

We are proving a statement of the form $\forall G.\ P(G)$, where

$$P(G) = \text{"}G \text{ has two disjoint homogeneous 4-sets"}.$$

The key fact: complementing a graph does not change which sets are homogeneous. A clique in G becomes an independent set in $\bar{G}$, and vice versa — but "homogeneous" means *clique or independent set*, so the collection of homogeneous sets is literally identical. Therefore

$$P(G) \iff P(\bar{G}).$$

The argument. Take an arbitrary G. Ramsey hands us a homogeneous 4-set A. Two cases:

- A is a clique in G: run the main argument on G.
- A is independent in G: then A is a **clique in** $\bar{G}$. Run the main argument on $\bar{G}$ to get $P(\bar{G})$. Since $P(\bar{G}) \iff P(G)$, we have proved $P(G)$ anyway.

So we never lose generality. The relabelling to $A = \{0, 1, 2, 3\}$ is free for the same reason: P is isomorphism-invariant.

Insight. The WLOG is licensed by an invariance of *the property being proved*, not by a symmetry of some particular graph. That is the general test: a "WLOG by complementation" is valid exactly when your target property is complement-invariant. Had we been proving something about edge counts, the same move would be illegal.

9 Part 9: The SAT encoding, built by hand

Let us construct actual conditions and an actual clause, so you can see there is no magic.

Setup. $A = \{a_0, a_1, a_2, a_3\}$ is a clique. $B = \{b_0, \dots, b_{15}\}$ carries one of the two fixed cores. The 64 variables are

$$x[i][j] = \text{“is } a_i \text{ joined to } b_j\text{?”}$$

Everything else is already determined.

9.1 Case 3-in-A, 1-in-B: take $\{a_0, a_1, a_2, b_5\}$

- *Clique?* The pairs a_0a_1, a_0a_2, a_1a_2 are already joined (A is a clique). We need the three cross pairs too. So: clique $\iff x[0][5] \wedge x[1][5] \wedge x[2][5]$.
- *Independent?* Impossible — a_0a_1 is an edge.

So this set is homogeneous under exactly one 3-literal condition. Count: 4 choices of triple from A, times 16 choices of b, $= 64$.

9.2 Case 2-in-A, 2-in-B: take $\{a_0, a_1, b_3, b_7\}$

- *Clique?* Need a_0a_1 (yes), need b_3b_7 **to be an edge of the core** (fixed — it either is or is not), plus all four cross edges. So *if* b_3b_7 is a core edge: $x[0][3] \wedge x[0][7] \wedge x[1][3] \wedge x[1][7]$, four literals. If it is not a core edge, this set can never be a clique.
- *Independent?* Impossible — a_0a_1 again.

Count: $\binom{4}{2} = 6$ pairs, times 60 core edges, $= 360$.

9.3 Case 1-in-A, 3-in-B: take $\{a_0, b_1, b_4, b_9\}$

Now both types are possible, depending on the core:

- *Clique*, if $b_1b_4b_9$ is a **triangle** in the core: $x[0][1] \wedge x[0][4] \wedge x[0][9]$.
- *Independent*, if $b_1b_4b_9$ is an **independent triple** in the core: $\neg x[0][1] \wedge \neg x[0][4] \wedge \neg x[0][9]$.

Count: 4×56 triangles $= 224$, plus 4×56 independent triples $= 224$.

9.4 The two excluded cases

- **All four in** B — impossible, the core is hom-4-free.
- A **itself** — it *is* a clique, but any set disjoint from A lies entirely in B, and B has no homogeneous 4-set. So A can never be half of a disjoint pair.

Total: $64 + 360 + 224 + 224 = 872$**.**

9.5 An actual clause

Take $P = \{a_0, a_1, a_2, b_5\}$ and $Q = \{a_3, b_1, b_4, b_9\}$ (with $b_1 b_4 b_9$ a core triangle). These are **vertex-disjoint**. So we forbid both firing at once:

$$\neg x[0][5] \vee \neg x[1][5] \vee \neg x[2][5] \vee \neg x[3][1] \vee \neg x[3][4] \vee \neg x[3][9]$$

A 6-literal clause. Do that for every disjoint pair, giving $104{,}524$ clauses.

9.6 A check you can do in your head

Clause length is the sum of the two conditions' literal counts. Which combinations are possible?

pairing	A-vertices used	length
(3A+1B) with (1A+3B)	$3 + 1 = 4$, fine	$3 + 3 = 6$
(1A+3B) with (1A+3B)	$1 + 1 = 2$, fine	$3 + 3 = 6$
(2A+2B) with (1A+3B)	$2 + 1 = 3$, fine	$4 + 3 = 7$
(2A+2B) with (2A+2B)	$2 + 2 = 4$, fine	$4 + 4 = 8$
(3A+1B) with (3A+1B)	$3 + 3 = 6 > 4$	impossible

So lengths can only be 6, 7, 8 — and the reported histogram is exactly $\{6 : 43{,}840,\ 7 : 52{,}416,\ 8 : 8{,}268\}$. That is a consistency check derived from first principles, and the sort of thing that would break loudly if the encoding were wrong.

A satisfying assignment would be a wiring of the 64 cross edges producing a graph with no two disjoint homogeneous 4-sets. Both instances are UNSAT: no such wiring exists.

> **Insight.** The honest epistemics: you do not "understand" an UNSAT result the way you understand the pigeonhole argument. What you can check is (i) that the encoding faithfully says what it claims, and (ii) that the refutation certificate is valid. Both were done. That is the substitute for intuition here, and it is a normal situation in modern combinatorics — the four-colour theorem has the same character.

10 Part 10: Why exactly two 16-vertex cores

The engine. Build hom-4-free graphs one vertex at a time.

Why it is exhaustive — this is the crux:

> Every induced subgraph of a hom-4-free graph is hom-4-free.

(Obvious once stated: a homogeneous 4-set in the subgraph is one in the whole graph.)

So if H is a hom-4-free graph on $n+1$ vertices, delete any vertex — you get a hom-4-free graph on n vertices. Meaning **every** level-$(n+1)$ graph arises from **some** level-n graph by adding a vertex. If your level-n list is complete, and you try every admissible neighbourhood, your level-$(n+1)$ list is complete. Induct from the trivial level 1.

When is adding vertex v with neighbourhood S admissible? A new homogeneous 4-set must contain v, so it is $\{v\} \cup T$ for a triple T. Two ways to go wrong:

- $\{v\} \cup T$ is a $K_4 \iff T \subseteq S$ and T is a **triangle**. So: *no triangle inside S.*
- $\{v\} \cup T$ is independent $\iff T \subseteq V \setminus S$ and T is an **independent triple**. So: *no independent triple outside S.*

And that bounds the search. Since G is hom-4-free, $\omega(G) \leq 3$ and $\alpha(G) \leq 3$. Then $G[S]$ is triangle-free with $\alpha \leq 3$, so by $R(3,4) = 9$ we get $|S| \leq 8$. Symmetrically $|V \setminus S| \leq 8$. So

$$n - 8 \;\leq\; |S| \;\leq\; 8,$$

a huge pruning that makes the whole thing run in seconds.

The output:

n	1	2	3	4	5	6	7	8	9
count	1	2	4	9	24	84	362	2079	14701

n	10	11	12	13	14	15	**16**	17	**18**
count	103706	546356	1449166	1184231	130816	640	**2**	1	**0**

> **Insight.** The **0 at $n = 18$ is $R(4,4) = 18$ falling out as a by-product** — "no hom-4-free graph on 18 vertices" is exactly Ramsey's statement. So the computation that establishes the catalogue also re-derives the theorem it depends on. If the code were buggy you would need a conspiracy: matching the published counts at every level, *and* landing on exactly 0 at 18, *and* exactly 2 at 16. That is the strongest internal evidence in the whole proof.

11 Part 11: Paley(17), and a pleasing coincidence

Vertices are $\mathbb{Z}_{17}$; join i and j when $i - j$ is a nonzero **quadratic residue** mod 17 — the squares $\{1, 2, 4, 8, 9, 13, 15, 16\}$.

It is a well-defined graph because $-1 \equiv 16$ is itself a residue, so the relation is symmetric.

It is self-complementary. Multiply every vertex by a *non*-residue: residues map to non-residues, so edges map to non-edges. Hence Paley(17) is isomorphic to its own complement — which immediately gives $\omega = \alpha$.

And that common value is 3. $\omega \geq 3$ is easy: $\{0, 1, 2\}$ works, since $1-0 = 1$, $2-1 = 1$, $2-0 = 2$ are all residues. That $\omega \leq 3$ — no K_4 — is the classical fact, and it is exactly *why* $R(4, 4) > 17$. In our setting it also drops out of the generation: Paley(17) is the unique survivor at level 17.

> **Insight.** Paley(17) is the standard lower-bound construction for $R(4, 4) = 18$, *and* it is our lower-bound witness for $z(20) \geq 6$. Same graph, both jobs — because both need the same property: *every homogeneous set is small*. Ramsey wants no homogeneous 4-set; we want groups of size at most 3, so that 17 vertices cannot fit in 5 groups.

12 Part 12: Self-test

If you can answer these unaided, you own the argument.

1. Why does the proof not need to consider the case where A (the first homogeneous 4-set) is an independent set?
2. In the 2-in-A, 2-in-B case, why does the *core* edge b_3b_7 matter, but the A edge a_0a_1 not?
3. Why can a (3A+1B) condition never pair with another (3A+1B) condition?
4. Where exactly does $R(4,4) = 18$ get used — and where does the number 12 in "$z(12) \leq 4$" come from?
5. Why does adding three isolated vertices to Paley(17) not reduce its cochromatic number?
6. What would it take to make the whole thing collapse — that is, which single computational result, if wrong, breaks the proof?

13 Part 13: The artifacts

Everything referenced here lives in `~/z20_repro/`:

item	path
orderly generator (C + nauty)	`nauty/gen44.c`
cocolourability decision procedure	`nauty/cochk.c`
the two CNFs and DRUP certificates	`certificate/`
independent CNF regeneration + RUP checker	`verify_certificate.py`
`drat-trim` source, binary, regenerated CNFs, LRAT	`drat/`
`cake_lpr` (formally verified checker)	`cake_lpr/`
exact c_6 counter (subset DP)	`c6_fast.py`
result logs	`nauty/*_result.txt`
formal proof paper	`../z20_proof_paper.md`
Lean formalisation blueprint	`lean_blueprint.md`

Provenance. Every mathematical step was produced by AI systems: a Nous Hermes agent designed the original (unsuccessful) search harness driving GPT-5.6 Sol; GPT-5.6 Sol contributed the structural reduction and the certificates; Claude carried out the independent verification, the catalogue and Ramsey recomputation, the $z(8)$ and $z(12)$ re-derivations, the lower bound, and this document. The result should be treated as a candidate proof pending independent scrutiny.